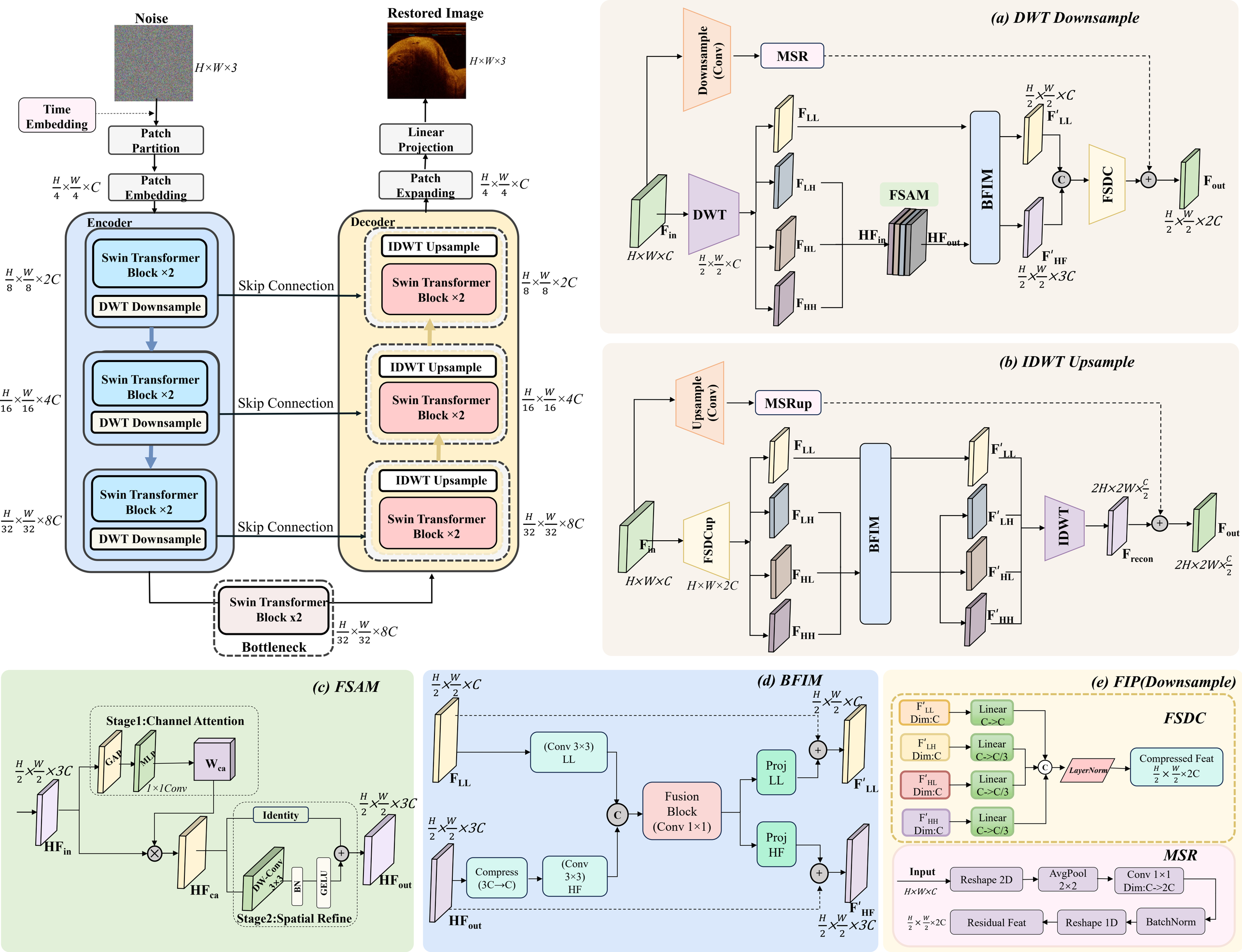
Noise
H×W×3
Time Embedding
Patch Partition
Patch Embedding
H/4×W/4×C
Encoder
Swin Transformer Block ×2
DWT Downsample
H/8×W/8×2C
H/16×W/16×4C
H/32×W/32×8C
Skip Connection
Swin Transformer Block x2
Bottleneck
Restored Image
Linear Projection
Patch Expanding
Decoder
IDWT Upsample
(a) DWT Downsample
Downsample (Conv)
MSR
DWT
FSAM
BFIM
FSDC
(b) IDWT Upsample
Upsample (Conv)
MSRup
FSDCup
IDWT
(c) FSAM
Stage1:Channel Attention
GAP
MLP
1×1Conv
Identity
DW-Conv 3×3
BN
GELU
Stage2:Spatial Refine
(d) BFIM
(Conv 3×3) LL
Compress (3C→C)
(Conv 3×3) HF
Fusion Block (Conv 1×1)
Proj LL
Proj HF
(e) FIP(Downsample)
FSDC
Linear C->C
Linear C->C/3
LayerNorm
Compressed Feat
MSR
Input
Reshape 2D
AvgPool 2×2
Conv 1×1 Dim:C->2C
BatchNorm
Reshape 1D
Residual Feat